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RATCHETING SETTING TOOL FOR AN EXPANDABLE SEAL IN A WELLBORE

Abstract

A ratcheting setting tool can use wellbore fluid to inflate expandable seals positioned downhole in a wellbore. For example, the ratcheting setting tool can include a piston within a first chamber. The piston can move from a first position to a second position in the first chamber when a pressure is applied to the piston. The ratcheting setting tool can include a first check valve in a second chamber. The first check valve can transmit wellbore fluid from the wellbore into the second chamber. The ratcheting setting tool can include a second check valve. The second check valve can transmit the wellbore fluid in the second chamber into an inflatable valve when the piston moves from the first position to the second position. In some examples, the ratcheting setting tool may be a hydraulic ratchet.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to wellbore operations and, more particularly (although not necessarily exclusively), to setting expandable seals in a wellbore.

BACKGROUND

[0002] During a completion phase of a hydrocarbon well, an open hole well zone (or interval) of interest may be isolated from a remainder of the hydrocarbon well for various reasons. Isolating an open hole well interval of interest is commonly accomplished using expandable, spaced apart seals. Expandable seals can be positioned on a tool string that can be used to run the expandable seals to a desired downhole location. After the expandable seals are positioned at the desired downhole location, the expandable seals may be expanded to isolate the well interval located between the expandable seals from fluid and pressure in other portions of the well.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 is a cross-sectional view of a wellbore environment including a ratcheting setting tool used to set expandable seals, according to an example of the present disclosure.

[0004] FIG. 2 is a cross-sectional view of an activation chamber for a hydraulic screen that is expanded by a ratcheting setting tool, according to an example of the present disclosure.

[0005] FIG. 3 is a cross-sectional diagram of a ratcheting setting tool in a first position, according to an example of the present disclosure.

[0006] FIG. 4 is a cross-sectional diagram of the ratcheting setting tool of FIG. 4 in a second position, according to an example of the present disclosure.

[0007] FIG. 5 is a flow chart representing a method for using a ratcheting setting tool to transfer wellbore fluid into an expandable seal in a wellbore, according to an example of the present disclosure.

DETAILED DESCRIPTION

[0008] Certain aspects and examples of the present disclosure relate to a ratcheting setting tool that can transfer wellbore fluid into downhole expandable seals such as packers or activation chambers for hydraulic screens. A ratcheting setting tool may be a hydraulic ratchet that can repeatedly actuate (e.g., in a linear direction) in successive pressure cycles to push wellbore fluid into an expandable seal. The ratcheting setting tool can include a piston that can be moved from a first position to a second position when a pressure is applied from a surface of a wellbore. When the piston moves to the second position, the piston can apply pressure to wellbore fluid in a chamber of the ratcheting setting tool. The wellbore fluid can enter the chamber via a first check valve to the annulus of the wellbore. When the piston applies pressure to the wellbore fluid, the wellbore fluid can be forced into an expandable seal via a second check valve, thus inflating the expandable seal. Because the second check valve can prevent the wellbore fluid from exiting the expandable seal, the pressure in the chamber can drop after the wellbore fluid is transferred. This drop in pressure can cause the piston to move back to the first position. The drop in pressure can also cause additional wellbore fluid to be transferred from the annulus and into the chamber. As the ratcheting setting tool is now reset, these operations (applying pressure from the surface, moving the piston, transferring the wellbore fluid into the expandable seal) can be repeated until the expandable seal is fully inflated with wellbore fluid.

[0009] Some expandable seals, such as packers, may require a considerable volume of fluid and relatively high pressures in order to fully inflate. For instance, some packers may require up to

8,000 psi to fully inflate. Some surface equipment may not be compatible with such high pressures. Other devices in a completion string, such as sleeves, may shear open at pressures less than 8,000 psi (e.g., before the packer is fully set), which may lead to well control issues. In some cases, cup tools may be used to isolate an inlet port of a packer. But this may not address issues related to the high pressures required to fully inflate the packer. Additionally, running a cup tool downhole may be time-consuming, as each packer on the completion string may need to be isolated in separate operations. In some cases, a completion string may have up to fifty packers and it may take several days to isolate all fifty packers.

[0010] Because the ratcheting setting tool described herein is resettable, the ratcheting setting tool can apply pressure cycles of a lower pressure (e.g., 2,000 psi) a number of times (e.g., in three cycles) to fully inflate a packer. Applying lower pressure cycles can prevent well control issues caused by high downhole pressures. Further, the ratcheting setting tool can be coupled to multiple packers downhole via a control line. Alternatively, each downhole device, such as but not limited to a packer or screen, may have its own ratcheting setting tool that can all be activated simultaneously by applying pressure through the tubing, the annulus, or both, as a tubing to annulus differential pressure need not be generated. Thus, the ratcheting setting tool may simultaneously inflate multiple packers, significantly reducing the amount of time needed to set packers downhole.

[0011] The ratcheting setting tool can similarly be used to set expandable hydraulic screens. As some hydraulic screens have a perforated base pipe, a tubing/annulus pressure differential may not be possible. Typically, slickline runs with cup packers can be run to set the hydraulic screens in groups of two, which can be time-consuming. Because the ratcheting setting tool can be directly attached to the expandable seals and may not depend on tubing to annulus differential for activation, the necessary pressure differential can be created that can allow the expandable seals to expand the hydraulic screens against the wellbore. Thus, by using the ratcheting setting tool described herein, a perforated base pipe with hydraulic screens can be deployed downhole as quickly as a solid base pipe with hydraulic screens.

[0012] Illustrative examples are given to introduce the reader to the general subject matter discussed herein and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative aspects, but, like the illustrative aspects, should not be used to limit the present disclosure.

[0013] FIG. 1 is a cross-sectional view of a wellbore environment **100** including a ratcheting setting tool used to set expandable seals, according to an example of the present disclosure. A floating workstation **102** (e.g., an oil platform or an offshore platform) can be centered over a submerged oil or gas well located in a sea floor **104** having a wellbore **106**. The wellbore **106** may extend from the sea floor **104** through a subterranean formation **108**. The subterranean formation **108** can include a fluid-bearing formation **110**. A subsea conduit **112** can extend from the deck **114** of the floating workstation **102** into a wellhead installation **116**. The floating workstation **102** can have a derrick **118** and a hoisting apparatus **120** for raising and lowering tools to drill, test, and complete the oil or gas well. The floating workstation **102** can be an oil platform as depicted in FIG. 1 or an aquatic vessel capable of performing the same or similar drilling and testing operations. In some examples, the process described herein can be applied to a land-based environment for wellbore exploration, planning, drilling, and completion.

[0014] A tubing string **122** can be lowered into the wellbore **106** of the oil or gas well as part of a completion operation of the oil or gas well. Downhole fluids, such as production fluids, can flow through a flow path defined by the tubing string **122**. The tubing string **122** can include one or more downhole tools usable downhole. The downhole tools can include wellbore stimulation equipment, production equipment, sand control tools, packers, retrievable tools, flow control devices, or any other suitable downhole tools. In the example depicted in FIG. 1, the downhole

tools can include expandable seal **123** (e.g., packers). The tubing string **122** can additionally include a ratcheting setting tool **125** that can be used to mechanically deform the expandable seal **123**. The ratcheting setting tool **125** can use a pressure multiplier (also sometimes known as a pressure intensifier) and piston travel to pump wellbore fluid into the expandable seals **123**. The wellbore fluid can include drilling fluid (e.g., drilling mud), brine, water, or any other suitable wellbore fluid. The ratcheting setting tool **125** may be a hydraulic ratchet that can pump the wellbore fluid into the expandable seals **123** over multiple cycles. By varying and cycling tubing pressure, a bicycle pump-like action can be achieved via the ratcheting setting tool **125** to inflate the expandable seals **123** over multiple cycles.

[0015] In some examples, the ratcheting setting tool **125** can simultaneously inflate multiple expandable seals **123** in a single operation. For example, the setting tool **125** can be coupled to each expandable seal **123** using a control line **101**. The ratcheting setting tool **125** can pump wellbore fluid through the control line **101** to multiple expandable seals **123** simultaneously (e.g., in a single pressure cycle). Alternatively, each downhole device can have its own ratcheting setting tool and pressure cycles through the tubing or annulus or both can activate each ratcheting setting tool simultaneously.

[0016] FIG. **2** is a cross-sectional view of an activation chamber **202** for a hydraulic screen **204** that is expanded by a ratcheting setting tool **125**, according to an example of the present disclosure. The activation chamber **202** can be an example of an expandable seal (e.g., the expandable seal **123** of FIG. **1**). When the activation chamber **202** is expanded, the activation chamber **202** can expand the hydraulic screen **204** (e.g., to conform against a wellbore **208**). In some examples, the activation chamber **202** and hydraulic screen **204** may be coupled to a perforated base pipe **206**. Perforations in the perforated base pipe **206** may prevent pressurization of fluid in the perforated base pipe **206**, preventing such fluid from inflating the activation chamber **202**. Typically, slickline runs may be performed to individually set isolation seals in order to inflate the activation chamber **202**. But this may be time consuming and labor intensive.

[0017] Instead, the ratcheting setting tool **125** can be used to draw in wellbore fluid (e.g., from an annulus **210** between the wellbore **208** and the perforated base pipe **206** or from the perforated base pipe **206**). As the perforated base pipe **206** is run downhole, wellbore fluid may be drawn into the activation chamber **202** to equalize a downhole pressure and a pressure inside the activation chamber **202**. After the perforated base pipe **206** has reached a desired position downhole, the ratcheting setting tool **125** can use pressure intensification (e.g., via a piston and in response to pressure applied from a surface of the wellbore **208**) to cause additional wellbore fluid to enter the activation chamber **202**. The ratcheting setting tool **125** can reset after each pressure cycle, allowing the activation chamber **202** to be expanded in multiple pressure cycles. The expanded activation chamber **202** can thus expand the hydraulic screen **204** against the wellbore **208**. The reset mechanism after each pressure cycle may ensure that the ratcheting setting tool **125** need not be long enough to carry the entire volume of fluid needed for activation in a single stroke of the ratcheting setting tool **125**.

[0018] FIG. **3** is a cross-sectional diagram of a ratcheting setting tool **125** in a first position, according to an example of the present disclosure. The ratcheting setting tool **125** can be attached to an expandable seal **123** (e.g., a packer, the activation chamber **202** of FIG. **2**, or any suitable inflatable element used downhole in a wellbore). The ratcheting setting tool **125** and the expandable seal **123** can be attached to a tubing string **122** in a wellbore **106**.

[0019] The ratcheting setting tool **125** can include a first chamber **302** and a second chamber **304**. The first chamber **302** may have a larger diameter than the second chamber **304**. This larger diameter can be used to intensify a pressure that is applied from a surface of the wellbore **106**. The first chamber **302** may have a first port **306** that can receive the pressure applied from the surface of the wellbore **106**. The ratcheting setting tool **125** can include a piston **308** within the first chamber **302**. In some examples, the ratcheting setting tool **125** can include one or more seals **310**

(e.g., an O-ring) between the piston **308** and the walls of the first chamber **302**. The seals **310** can prevent the surface pressure received via the first port **306** from leaking through the first chamber **302** and into the second chamber **304**.

[0020] The piston **308** can include an extension piece **312** that can extend into the second chamber **304**. In some examples, the ratcheting setting tool **125** can include one or more seals **314** (e.g., O-rings) between the extension piece **312** and the walls of the second chamber **304**. The seals **314** can prevent a pressure in a volume **316** between the seals **314** and the piston **308** from leaking into a remainder of the second chamber **304**. In some examples, the piston **308** and extension piece **312** can contain a nitrogen gas spring. That is, the volume **316** may include a compressible gas or fluid such as nitrogen. In other examples, the piston **308** and extension piece **312** may support a compressible mechanical spring.

[0021] In some examples, the second chamber **304** can include an annulus port **318** to an annulus **322** between the tubing string **122** and the wellbore **106**. In other examples, the second chamber **304** can include an annulus port **318** to the tubing string **122**. The annulus **322** and the tubing string **122** can include wellbore fluid (e.g., drilling mud pumped from a surface of the wellbore **106**, water, hydrocarbons, or any suitable wellbore fluid). A first check valve **320** can be positioned in the annulus port **318**. The first check valve **320** can allow wellbore fluid (e.g., from the annulus **322** or from the tubing string **122**) to enter the second chamber **304** via the annulus port **318**. But the first check valve **320** may not allow wellbore fluid to exit the second chamber **304** via the annulus port **318**. In some examples, the annulus port **318** can include a screen **319** or a mechanical filter that can cleanse the wellbore fluid before the wellbore fluid enters the first check valve **320**. The screen **319** can filter out relatively large particles that may otherwise get caught in the first check valve **320**. Additionally or alternatively, the ratcheting setting tool **125** may include an additional fluid chamber (e.g., coupled to the first check valve **320**) that can transfer wellbore fluid chamber into the second chamber **304**. For example, a control line (e.g., the control line **101** of FIG. 1) may transfer wellbore fluid into the additional fluid chamber.

[0022] The second chamber **304** can also include a second port **324** that couples the ratcheting setting tool **125** to the expandable seal **123**. A second check valve **326** can be positioned in the second port **324**. The second check valve **326** can allow wellbore fluid in the second chamber **304** to enter the expandable seal **123** via the second port **324**. But the second check valve **326** may not allow wellbore fluid to exit the expandable seal **123** via the second port **324**. Thus, wellbore fluid can be transferred into the expandable seal **123** and can remain in the expandable seal **123** to cause an expansion of the expandable seal **123**. In some examples, the first check valve **320** and the second check valve **326** can include a ceramic ball and a polytetrafluoroethylene (PTFE) seat that can seal the ceramic ball.

[0023] In some examples, the ratcheting setting tool **125** may include one or more rupture disks. For example, the ratcheting setting tool **125** may include a first rupture disk **332** on the first port **306** of the first chamber **302**. The first rupture disk **332** can rupture if downhole pressure exceeds a rupture threshold. In some examples, the tubing string **122** may be moved up and down or in and out of the wellbore **106** several times in the process of setting the tubing string **122** downhole. In some examples, this may increase a downhole pressure applied to the ratcheting setting tool **125**, causing the ratcheting setting tool **125** to prematurely transfer wellbore fluid into the expandable seal **123** to expand the expandable seal **123**. Placing the first rupture disk **332** on the first port **306** can prevent the ratcheting setting tool **125** from premature operations. When the tubing string **122** is set downhole, a pressure can be applied from the surface of the wellbore **106** that can rupture the first rupture disk **332**, thus allowing the ratcheting setting tool **125** to be used to expand the expandable seal **123**.

[0024] Additionally, in some examples, the second chamber **304** may have a second annulus port **328** to the annulus **322** or the tubing string **122**. The second annulus port **328** may have a second rupture disk **330** that can rupture when pressure within the second chamber **304** exceeds a rupture

threshold. In some examples, the rupture threshold for the second rupture disk **330** may be higher than a rupture threshold for the first rupture disk **332**. The second rupture disk **330** may be beneficial for activation chambers used to expand hydraulic screens, as in some examples, such hydraulic screens may not have an activation limiter. Rupturing the second rupture disk **330** may cause wellbore fluid from the annulus **322** or the tubing string **122** to enter the second chamber **304**. This can cause the pressure inside the second chamber **304** to equalize with the pressure in the annulus **322** or the tubing string **122** permanently by opening a permanent flow path between the second chamber **304** and the tubing string **122** or annulus **322**. Hence, no further activation of the expandable seal **123** may be possible, thereby ensuring that the expandable seal **123** is not over pressured or otherwise over activated.

[0025] As depicted in FIG. 3, before operation, the piston **308** of the ratcheting setting tool **125** may be positioned in a first position. That is, the first position can involve the piston **308** being set at a distance away from a leftmost wall of the first chamber **302**. In a non-limiting example, the downhole pressure in the annulus **322** can be 3,000 psi. As the tubing string **122** is run downhole, the first check valve **320** can allow wellbore fluid to enter the second chamber **304** until a pressure inside the second chamber **304** equalizes to the pressure outside the second chamber **304** (e.g., 3,000 psi). Similarly, the second check valve **326** can allow the wellbore fluid inside the second chamber **304** to transfer into the expandable seal **123** until the pressure inside the expandable seal **123** equalizes (e.g., to 3,000 psi). This may not yet cause an inflation of the expandable seal **123**, as the pressure inside and outside of the expandable seal **123** may be equal. The pressure in the first chamber **302** (e.g., to the right of the piston **308**) can also be 3,000 psi. The pressure in the volume **316** between the seals **314** and the piston **308**, which may include a compressible gas or fluid (e.g., nitrogen) or a mechanical spring, can also be 3,000 psi. Pressure can be applied from uphole to actuate the ratcheting setting tool **125** to inflate the expandable seal **123**.

[0026] For example, upon the tubing string **122** reaching a desired depth, the pressure for the tubing string **122** and annulus **322** can be increased by 500 psi. Because there is not a need to create a pressure differential between the tubing string **122** and the annulus **322**, the pressure can be applied across the entire wellbore (e.g., including within the tubing string **122**). Thus, the pressure in the annulus **322** can be 3,500 psi and the pressure in the first chamber **302** can be 3,500 psi. The cross-sectional area of the first chamber **302** can be greater than the cross-sectional area of the second chamber **304**. For example, the cross-sectional area of the first chamber **302** may be double the cross-sectional area of the second chamber **304**. This difference in areas can cause a pressure multiplier effect on the wellbore fluid in the second chamber **304**. For example, the additional 500 psi of pressure applied downhole and inside the first chamber **302** can be converted to 1,000 psi of pressure applied to the left side of the piston **308** and extension piece **312** (e.g., in the second chamber because the area of the first chamber **302** is double the area of the second chamber **304**). Additionally, the 500 psi of pressure applied to the piston **308** can move the piston **308** from the first position to a second position.

[0027] FIG. 4 is a cross-sectional diagram of the ratcheting setting tool of FIG. 4 in a second position, according to an example of the present disclosure. The second position can involve the leftmost side of the piston **308** contacting the leftmost wall of the first chamber **302**. In other words, the entire extension piece **312** may be positioned within the second chamber **304** when the piston **308** is in the second position. Continuing the non-limiting example introduced above, the pressure within the second chamber **304** can be 4,000 psi due to the piston **308** moving to the second position and due to the pressure multiplier effect caused by the difference in cross-sectional area between the first chamber **302** and the second chamber **304**.

[0028] Increasing the pressure in the second chamber **304** to 4,000 psi while the pressure in the annulus **322** remains at 3,500 psi can cause the first check valve **320** to close. Closing the first check valve **320** can prevent wellbore fluid from the annulus **322** from flowing into the second chamber **304** via the annulus port **318**. As the pressure inside the second chamber **304** is now 4,000

psi while the pressure inside the expandable seal **123** is 3,000 psi, this pressure differential may cause the second check valve **326** to open. Opening the second check valve **326** can push the wellbore fluid in the second chamber **304** into the expandable seal **123** via the second port **324** until the pressure inside the expandable seal **123** equalizes with the pressure in the second chamber **304** (e.g., to 3,500 psi). Because the second check valve **326** can prevent wellbore fluid from flowing out of the expandable seal **123**, the additional wellbore fluid can cause the expandable seal **123** to expand.

[0029] Once wellbore fluid has been pushed into the expandable seal **123**, the pressure inside the second chamber **304** may be 3,500 psi. The 500 psi of pressure applied from uphole may cease. This can cause the pressure in the annulus **322**, the tubing string **122**, and the first chamber **302** to return to the initial downhole pressure (e.g., 3,000 psi). Because the pressure to the left of the piston **308** and extension piece **312** (e.g., 3,500 psi) is greater than the pressure to the right of the piston **308**, and because of energy stored in a mechanical spring or a compressible fluid spring in the volume **316**, the piston **308** can be pushed back to the first position.

[0030] Pushing the piston **308** back to the first position can cause a pressure drop in the second chamber **304**. The piston **308** may be pushed back to the first position due to the force from the gas spring in volume **316**, a mechanical spring in volume **316**, or due to a reduction in the downhole pressure. For example, the pressure in the second chamber **304** may drop to 2,500 psi. The pressure in the annulus may be 3,000 psi. This pressure differential can cause the first check valve **320** to open. The open first check valve **320** can allow additional wellbore fluid to enter the second chamber **304** from the annulus **322** via the annulus port **318** until the pressure inside the second chamber **304** equalizes with the pressure outside the second chamber **304** (e.g., to 3,000 psi). Thus, the ratcheting setting tool **125** can be fully reset to the first position to be used in a second pressure cycle (e.g., applying pressure from the surface of the wellbore **106** to transfer wellbore fluid into the expandable seal **123**). Pressure cycles can be repeated until the expandable seal **123** is fully inflated. For example, wellbore fluid may be transferred into the expandable seal **123** until there is a pressure differential of 750 psi between the expandable seal **123** and the annulus **322**. Thus, the expandable seal **123** may be inflated solely with surface equipment and without the use of a slickline rig or crew.

[0031] The number of pressure cycles and the amount of pressure applied from uphole in each pressure cycle can depend on the size of the ratcheting setting tool **125**, the size of the expandable seal **123**, and the number of expandable seals **123** connected to the ratcheting setting tool **125** (e.g., via a control line connected to inlet ports of the expandable seals **123**). If the tubing string **122** includes tools with pressure limits (e.g., sleeves) or if the wellbore cannot support high applied pressure (e.g., fracturing of the formation), then the uphole pressure may be relatively low (e.g., below the pressure limits), and an increased number of pressure cycles may be performed to fully inflate the expandable seal **123**. Alternatively, a larger area ratio between first chamber **302** and second chamber **304** may be utilized. As a non-limiting example, a ratio of 4:1 can intensify the applied pressure by 4 times. One skilled in the art will recognize that the ratio of areas and the size of the tool can be modified to suit different applications and requirements without departing from the scope of the invention.

[0032] FIG. 5 is a flow chart representing a process **500** for using a ratcheting setting tool **125** to transfer wellbore fluid into an expandable seal **123** in a wellbore **106**, according to an example of the present disclosure. The steps of process **500** may be described with reference to components discussed above with respect to FIGS. 1-4. Some of the following steps may be performed in any order with respect to the other steps as would be understood by one of ordinary skill in the art.

[0033] At block **502**, a ratcheting setting tool **125** positioned downhole in a wellbore **106** can receive a pressure (e.g., a first pressure) applied from a surface of the wellbore **106**. The ratcheting setting tool **125** can be a hydraulic ratchet. The ratcheting setting tool **125** can define a first chamber **302** and a second chamber **304**. The pressure can be received at a first port **306** of the first

chamber **302**. The first chamber **302** can include the first port **306** and a piston **308** with an extension piece **312** that can extend into the second chamber **304**. A second port **324** in the second chamber **304** can couple the ratcheting setting tool **125** to an expandable seal **123**. Prior to receiving the first pressure applied from the surface of the wellbore **106**, the internal pressure of the first chamber **302**, the second chamber **304**, and the expandable seal **123** may be equalized to a downhole pressure in an annulus **322** of the wellbore **106** external to the ratcheting setting tool **125** and the expandable seal **123**. Applying the first pressure from the surface of the wellbore **106** can cause a pressure within the first chamber **302** to be greater than a pressure within the second chamber **304**.

[0034] At block **504**, a piston **308** within the first chamber **302** defined by the ratcheting setting tool **125** can be moved from a first position to a second position in response to receiving the first pressure. Moving from the first position to the second position may involve moving the piston **308** towards the second chamber **304**, thus applying a second pressure to increase a pressure within the second chamber **304**. In some examples, a cross-sectional area of the first chamber **302** may be larger than a cross-sectional area of the second chamber **304**. This can cause a pressure multiplier effect on the first pressure applied from the surface of the wellbore **106**. For example, if the first pressure applied from the surface is 500 psi and the cross-sectional area of the first chamber **302** is double the cross-sectional area of the second chamber **304**, moving the piston **308** from the first position to the second position may cause double the amount of first pressure from the surface to be applied to the second chamber **304** (e.g., 1,000 psi as a second pressure).

[0035] At block **506**, a first check valve **320** of the second chamber **304** defined by the ratcheting setting tool **125** can transmit wellbore fluid from the annulus **322** of the wellbore **106** into the second chamber **304**. The first check valve **320** may be positioned in an annulus port **318** of the second chamber **304**. The annulus port **318** may be a port to the annulus **322** or to the tubing string **122**. The first check valve **320** may allow wellbore fluid to flow into the second chamber **304** but may prevent wellbore fluid from flowing out of the second chamber **304**. Thus, when a pressure inside the second chamber **304** is less than a downhole pressure in the wellbore **106**, the first check valve **320** may open and wellbore fluid can flow into the second chamber **304**. When the pressure inside the second chamber **304** is greater than a downhole pressure in the wellbore **106**, the first check valve **320** may close and wellbore fluid may not flow into the second chamber **304**.

[0036] At block **508**, a second check valve **326** in the ratcheting setting tool **125** can transmit the wellbore fluid from the second chamber **304** into the expandable seal **123** in response to the piston **308** moving from the first position to the second position. The second check valve **326** can be positioned within the second port **324** that couples the ratcheting setting tool **125** to the expandable seal **123**. Moving the piston **308** to the second position can apply a second pressure to the second chamber **304**, thus increasing the pressure. The second chamber **304** can then have a higher internal pressure than a pressure within the expandable seal **123**. This pressure differential can cause the second check valve **326** to open, allowing the wellbore in the second chamber **304** to transfer into the expandable seal **123**. The second check valve **326** may not allow wellbore fluid to transfer out of the expandable seal **123**. Thus, the wellbore fluid transferred into the expandable seal **123** in response to the piston **308** moving to the second position may remain in the expandable seal **123**.

[0037] In some examples, after the second check valve **326** transmits the wellbore fluid into the expandable seal **123**, the piston **308** can move back to the first position from the second position. The piston **308** may include a spring storing energy that causes the piston **308** to move back to the first position. When the piston **308** moves back to the first position, this can create a vacuum effect in the second chamber **304**. For example, the pressure within the second chamber **304** can drop to be lower than the downhole pressure in the annulus **322** or the tubing string **122**. This can cause the first check valve **320** to open, allowing additional wellbore fluid from the annulus **322** or the tubing string **122** to be transmitted into the second chamber **304**. The ratcheting setting tool **125** can therefore be reset to an initial position to be used in further pressure cycles.

[0038] In some examples, at least three pressure cycles may be performed by the ratcheting setting tool **125** to mechanically deform (e.g., inflate or expand) the expandable seal **123**, as a single cycle may be insufficient to fully set the expandable seal **123**. In some examples, up to fifty pressure cycles may be performed by the ratcheting setting tool **125** to set the expandable seal **123**. For example, the ratcheting setting tool **125** may perform up to five, up to ten, up to fifteen, up to twenty, up to twenty-five, up to thirty, up to thirty-five, up to forty, up to forty-five, or up to fifty pressure cycles to set the expandable seal **123**. In some examples, performance of a single pressure cycle by the ratcheting setting tool **125** may take between one minute and fifty minutes. For example, a single pressure cycle may take between one minute and five minutes, between five minutes and ten minutes, between ten minutes and fifteen minutes, between fifteen minutes and twenty minutes, between twenty minutes and twenty-five minutes, between twenty-five minutes and thirty minutes, between thirty minutes and thirty-five minutes, between thirty-five minutes and forty minutes, or between forty-five minutes and fifty minutes.

[0039] In some examples, the ratcheting setting tool **125** may be coupled to multiple expandable seals (e.g., packers or activation chambers) via a control line **101**. The control line **101** can transmit wellbore fluid from the second chamber **304** into the additional packers or activation chambers. The ratcheting setting tool **125** can therefore be used to simultaneously inflate multiple packers or activation chambers at the same time.

[0040] In some aspects, system, method, and apparatus for transferring wellbore fluid into downhole expandable seals are provided according to one or more of the following examples:

[0041] As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., “Examples 1-4” is to be understood as “Examples 1, 2, 3, or 4”).

[0042] Example 1 is a system comprising: an expandable seal positionable downhole in a wellbore; and a ratcheting setting tool comprising: a piston positionable within a first chamber defined by the ratcheting setting tool, the piston movable from a first position to a second position within the first chamber in response to a pressure; a first check valve configured to transmit wellbore fluid from the wellbore into a second chamber defined by the ratcheting setting tool; and a second check valve configured to transmit wellbore fluid from the second chamber into the expandable seal in response to the piston moving from the first position to the second position.

[0043] Example 2 is the system of example(s) 1, wherein the ratcheting setting tool further comprises: an extension piece extending from the piston into the second chamber; a first seal coupleable to the piston in the first chamber; and a second seal coupleable to the extension piece in the second chamber, wherein the ratcheting setting tool is a hydraulic ratchet.

[0044] Example 3 is the system of any of example(s) 1-2, wherein the expandable seal is a first expandable seal on a tubing string deployable downhole in the wellbore, and wherein the system further comprises: a control line coupleable at a first end to a third port of the second chamber and at a second end to an inlet port of a second inflatable packer, the control line configurable to transfer wellbore fluid into the second inflatable packer via the inlet port.

[0045] Example 4 is the system of any of example(s) 1-3, wherein the expandable seal is positionable between an outer surface of a perforated tubing string and a hydraulic screen, and wherein the expandable seal is configurable to expand the hydraulic screen in the wellbore.

[0046] Example 5 is the system of any of example(s) 1-4, wherein the ratcheting setting tool further comprises: a first port in the first chamber configured to receive the pressure; an annulus port in the second chamber, wherein the first check valve is positioned in the annulus port; and a second port in the second chamber coupleable to the expandable seal, wherein the second check valve is positioned in the second port.

[0047] Example 6 is the system of any of example(s) 1-5, further comprising: a first rupture disk positionable on the first port of the ratcheting setting tool.

[0048] Example 7 is the system of any of example(s) 1-6, wherein the annulus port is a first

annulus port, and wherein the system further comprises: a second rupture disk positionable on a second annulus port of the second chamber.

[0049] Example 8 is a method comprising: receiving, by a ratcheting setting tool positioned downhole in a wellbore, a pressure; moving a piston within a first chamber defined by the ratcheting setting tool from a first position to a second position in response to receiving the pressure; transmitting, by a first check valve in a second chamber defined by the ratcheting setting tool, wellbore fluid from the wellbore into the second chamber; and transmitting, by a second check valve in the ratcheting setting tool, the wellbore fluid from the second chamber into an expandable seal in response to the piston moving from the first position to the second position.

[0050] Example 9 is the method of example(s) 8, wherein the ratcheting setting tool is a hydraulic ratchet, wherein the pressure is a first pressure, and wherein the method further comprises: applying, by the piston, a second pressure to the wellbore fluid in the second chamber in response to the first pressure being applied from a surface of the wellbore, wherein the second pressure is greater than the first pressure.

[0051] Example 10 is the method of any of example(s) 8-9, further comprising: moving, by the piston, from the second position to the first position subsequent to the second check valve transmitting the wellbore fluid from the second chamber into the expandable seal; and transmitting, by the first check valve, additional wellbore fluid into the second chamber in response to the piston moving from the second position to the first position.

[0052] Example 11 is the method of any of example(s) 8-10, wherein the expandable seal is a first expandable seal on a tubing string deployed downhole in the wellbore, and wherein the method further comprises: transmitting, by a control line, the wellbore fluid from the second chamber of the ratcheting setting tool and into a second expandable seal.

[0053] Example 12 is the method of any of example(s) 8-11, further comprising: expanding the expandable seal in response to transmitting the wellbore fluid from the second chamber into the expandable seal; and expanding a hydraulic screen surrounding the expandable seal against the wellbore in response to expanding the expandable seal.

[0054] Example 13 is the method of any of example(s) 8-12, further comprising: rupturing a first rupture disk positioned on a first port in the first chamber in response to a downhole pressure of the wellbore exceeding a rupture threshold.

[0055] Example 14 is the method of any of example(s) 8-13, further comprising: rupturing a second rupture disk on a second port in the second chamber in response to a downhole pressure of the wellbore exceeding a rupture threshold.

[0056] Example 15 is a ratcheting setting tool comprising: a piston positionable within a first chamber defined by the ratcheting setting tool that is positionable downhole in a wellbore, the piston movable from a first position to a second position within the first chamber in response to a pressure; a first check valve configured to transmit wellbore fluid from the wellbore into a second chamber defined by the ratcheting setting tool; and a second check valve configured to transmit wellbore fluid from the second chamber into an expandable seal in response to the piston moving from the first position to the second position.

[0057] Example 16 is the ratcheting setting tool of example(s) 15, further comprising: an extension piece extending from the piston into the second chamber; a first seal coupleable to the piston in the first chamber; and a second seal coupleable to the extension piece in the second chamber.

[0058] Example 17 is the ratcheting setting tool of any of example(s) 15-16, further comprising: a first rupture disk positionable on a first port in the first chamber.

[0059] Example 18 is the ratcheting setting tool of any of example(s) 15-17, further comprising: a second rupture disk positionable on a second port of the second chamber.

[0060] Example 19 is the ratcheting setting tool of any of example(s) 15-18, wherein the expandable seal is a first expandable seal on a tubing string deployable downhole in the wellbore, and wherein the ratcheting setting tool is coupleable to a control line configurable to transmit

wellbore fluid from a third port of the second chamber and into a second inflatable packer via an inlet port.

[0061] Example 20 is the ratcheting setting tool of any of example(s) 15-19, further comprising: a first port in the first chamber configured to receive the pressure; an annulus port in the second chamber, wherein the first check valve is positioned in the annulus port; and a second port in the second chamber coupleable to the expandable seal, wherein the second check valve is positioned in the second port.

[0062] The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure.

Claims

1. A system comprising: an expandable seal positionable downhole in a wellbore; and a ratcheting setting tool comprising: a piston positionable within a first chamber defined by the ratcheting setting tool, the piston movable from a first position to a second position within the first chamber in response to a pressure; a first check valve configured to transmit wellbore fluid from the wellbore into a second chamber defined by the ratcheting setting tool; and a second check valve configured to transmit wellbore fluid from the second chamber into the expandable seal in response to the piston moving from the first position to the second position.
2. The system of claim 1, wherein the ratcheting setting tool further comprises: an extension piece extending from the piston into the second chamber; a first seal coupleable to the piston in the first chamber; and a second seal coupleable to the extension piece in the second chamber, wherein the ratcheting setting tool is a hydraulic ratchet.
3. The system of claim 1, wherein the expandable seal is a first expandable seal on a tubing string deployable downhole in the wellbore, and wherein the system further comprises: a control line coupleable at a first end to a third port of the second chamber and at a second end to an inlet port of a second inflatable packer, the control line configurable to transfer wellbore fluid into the second inflatable packer via the inlet port.
4. The system of claim 1, wherein the expandable seal is positionable between an outer surface of a perforated tubing string and a hydraulic screen, and wherein the expandable seal is configurable to expand the hydraulic screen in the wellbore.
5. The system of claim 1, wherein the ratcheting setting tool further comprises: a first port in the first chamber configured to receive the pressure; an annulus port in the second chamber, wherein the first check valve is positioned in the annulus port; and a second port in the second chamber coupleable to the expandable seal, wherein the second check valve is positioned in the second port.
6. The system of claim 5, further comprising: a first rupture disk positionable on the first port of the ratcheting setting tool.
7. The system of claim 5, wherein the annulus port is a first annulus port, and wherein the system further comprises: a second rupture disk positionable on a second annulus port of the second chamber.
8. A method comprising: receiving, by a ratcheting setting tool positioned downhole in a wellbore, a pressure; moving a piston within a first chamber defined by the ratcheting setting tool from a first position to a second position in response to receiving the pressure; transmitting, by a first check valve in a second chamber defined by the ratcheting setting tool, wellbore fluid from the wellbore into the second chamber; and transmitting, by a second check valve in the ratcheting setting tool, the wellbore fluid from the second chamber into an expandable seal in response to the piston moving from the first position to the second position.

- 9.** The method of claim 8, wherein the ratcheting setting tool is a hydraulic ratchet, wherein the pressure is a first pressure, and wherein the method further comprises: applying, by the piston, a second pressure to the wellbore fluid in the second chamber in response to the first pressure being applied from a surface of the wellbore, wherein the second pressure is greater than the first pressure.
- 10.** The method of claim 8, further comprising: moving, by the piston, from the second position to the first position subsequent to the second check valve transmitting the wellbore fluid from the second chamber into the expandable seal; and transmitting, by the first check valve, additional wellbore fluid into the second chamber in response to the piston moving from the second position to the first position.
- 11.** The method of claim 8, wherein the expandable seal is a first expandable seal on a tubing string deployed downhole in the wellbore, and wherein the method further comprises: transmitting, by a control line, the wellbore fluid from the second chamber of the ratcheting setting tool and into a second expandable seal.
- 12.** The method of claim 8, further comprising: expanding the expandable seal in response to transmitting the wellbore fluid from the second chamber into the expandable seal; and expanding a hydraulic screen surrounding the expandable seal against the wellbore in response to expanding the expandable seal.
- 13.** The method of claim 8, further comprising: rupturing a first rupture disk positioned on a first port in the first chamber in response to a downhole pressure of the wellbore exceeding a rupture threshold.
- 14.** The method of claim 8, further comprising: rupturing a second rupture disk on a second port in the second chamber in response to a downhole pressure of the wellbore exceeding a rupture threshold.
- 15.** A ratcheting setting tool comprising: a piston positionable within a first chamber defined by the ratcheting setting tool that is positionable downhole in a wellbore, the piston movable from a first position to a second position within the first chamber in response to a pressure; a first check valve configured to transmit wellbore fluid from the wellbore into a second chamber defined by the ratcheting setting tool; and a second check valve configured to transmit wellbore fluid from the second chamber into an expandable seal in response to the piston moving from the first position to the second position.
- 16.** The ratcheting setting tool of claim 15, further comprising: an extension piece extending from the piston into the second chamber; a first seal coupleable to the piston in the first chamber; and a second seal coupleable to the extension piece in the second chamber.
- 17.** The ratcheting setting tool of claim 15, further comprising: a first rupture disk positionable on a first port in the first chamber.
- 18.** The ratcheting setting tool of claim 15, further comprising: a second rupture disk positionable on a second port of the second chamber.
- 19.** The ratcheting setting tool of claim 15, wherein the expandable seal is a first expandable seal on a tubing string deployable downhole in the wellbore, and wherein the ratcheting setting tool is coupleable to a control line configurable to transmit wellbore fluid from a third port of the second chamber and into a second inflatable packer via an inlet port.
- 20.** The ratcheting setting tool of claim 15, further comprising: a first port in the first chamber configured to receive the pressure; an annulus port in the second chamber, wherein the first check valve is positioned in the annulus port; and a second port in the second chamber coupleable to the expandable seal, wherein the second check valve is positioned in the second port.
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